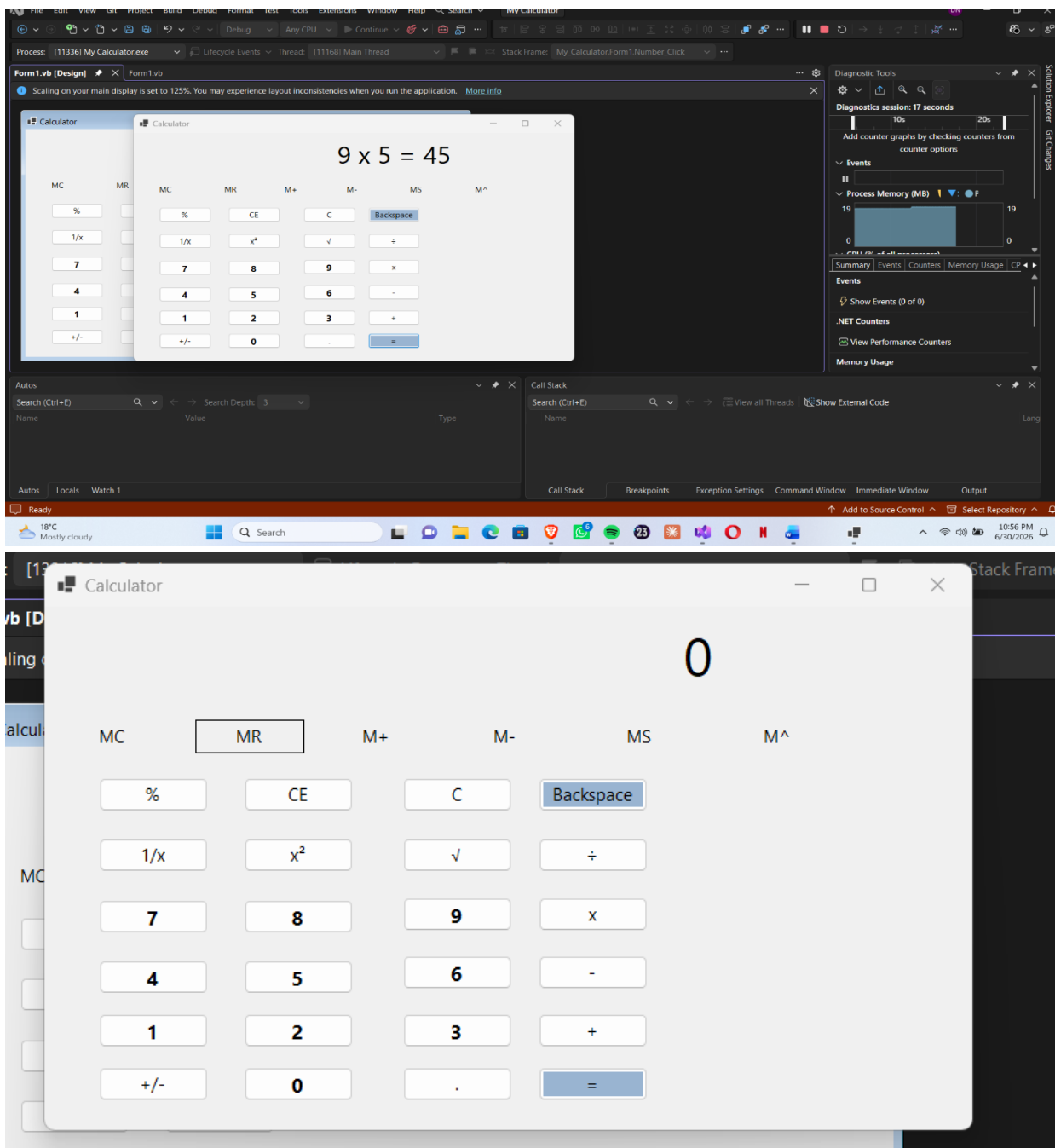


CAT 1 APPLICATION PROGRAMMING

Below are the various outputs for mathematical problems on my Calculator;



Visual Studio Code interface showing a debugging session for a Windows Calculator application. The main window displays the Calculator application with the calculation $10 \div 2 = 5$. The interface includes the Solution Explorer, Diagnostic Tools (showing a 1:17 minutes session), Autos, Call Stack, and various tool windows. The status bar indicates the application is running on the main thread.

Process: [11336] My Calculator.exe
Thread: [11168] Main Thread
Stack Frame: My_Calculator.Form1.Number_Click

Form1.vb [Design] | Form1.vb

Scaling on your main display is set to 125%. You may experience layout inconsistencies when you run the application. [More info](#)

Calculator

MC MR MC MR M+ M- MS M^

% 1/x 7 4 1 +/-

CE x^2 sqrt +

8 5 6 -

2 3 +

0 . =

Backspace

Diagnostic Tools

Diagnostics session: 1:17 minutes

1:10min

Add counter graphs by checking counters from counter options

Events

Process Memory (MB) 26 26

Summary Events Counters Memory Usage CP

Events

Show Events (0 of 0)

.NET Counters

View Performance Counters

Memory Usage

Autos

Search (Ctrl+E)

Search Depth: 3

Name Value Type

Autos Locals Watch 1

Call Stack

Search (Ctrl+E)

View all Threads Show External Code

Call Stack Breakpoints Exception Settings Command Window Immediate Window Output

Ready

18°C Mostly cloudy

Search

10:57 PM 6/30/2026

Visual Studio Code interface showing a debugging session for a Windows Calculator application. The main window displays the Calculator application with the calculation $60 - 45 = 15$. The interface includes the Solution Explorer, Diagnostic Tools (showing a 3:13 minutes session), Autos, Call Stack, and various tool windows. The status bar indicates the application is running on the main thread.

Process: [11336] My Calculator.exe
Thread: [11168] Main Thread
Stack Frame: My_Calculator.Form1.Number_Click

Form1.vb [Design] | Form1.vb

Scaling on your main display is set to 125%. You may experience layout inconsistencies when you run the application. [More info](#)

Calculator

MC MR MC MR M+ M- MS M^

% 1/x 7 4 1 +/-

CE x^2 sqrt +

8 5 6 -

2 3 +

0 . =

Backspace

Diagnostic Tools

Diagnostics session: 3:13 minutes

3:00min 3:10min

Add counter graphs by checking counters from counter options

Events

Process Memory (MB) 26 26

Summary Events Counters Memory Usage CP

Events

Show Events (0 of 0)

.NET Counters

View Performance Counters

Memory Usage

Autos

Search (Ctrl+E)

Search Depth: 3

Name Value Type

Autos Locals Watch 1

Call Stack

Search (Ctrl+E)

View all Threads Show External Code

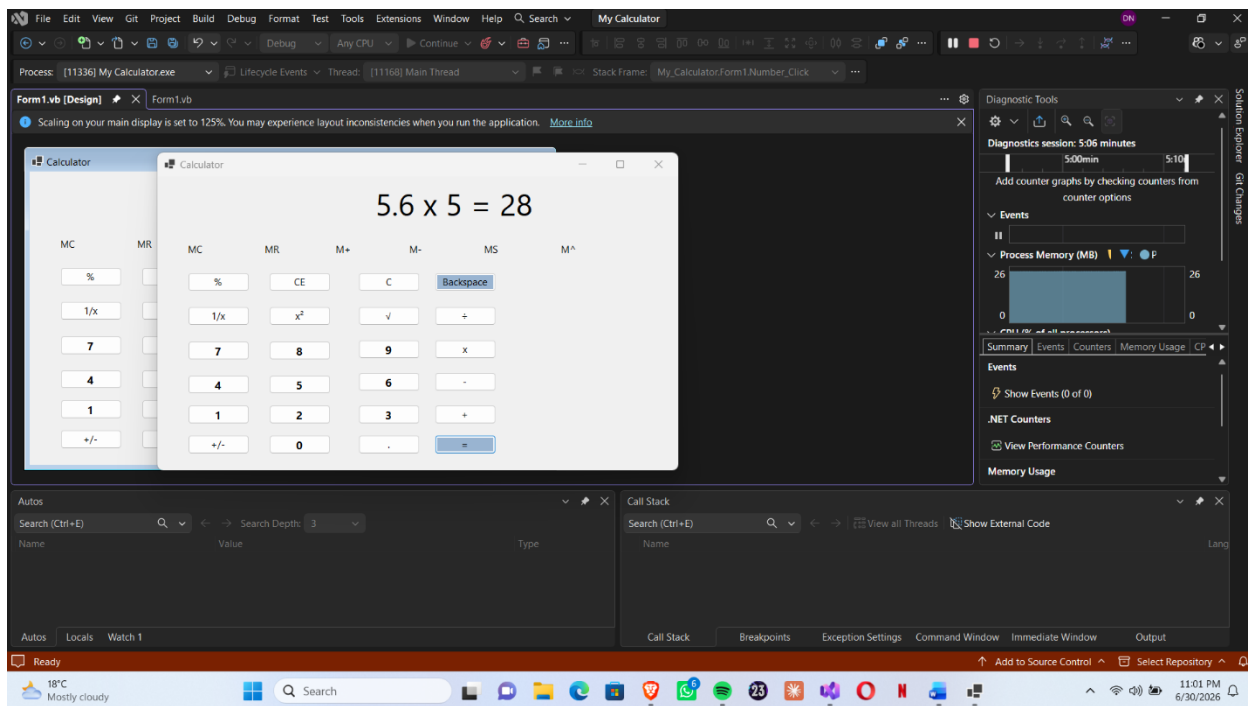
Call Stack Breakpoints Exception Settings Command Window Immediate Window Output

Ready

18°C Mostly cloudy

Search

10:59 PM 6/30/2026



Below are the code view snips for the Calculator Design;

Visual Studio Code interface showing the development of a Windows Forms application named "My Calculator". The main editor displays the code for `Form1.vb` in Design view, with the `Click` event handler for `btnEquals` selected. The code includes initialization of variables (`Dim operand1 As Double = 0`, `Dim currentOperator As String = ""`, `Dim isCalculationDone As Boolean = False`) and a `Form_Load` event handler. The `Click` event handler for `btnEquals` is currently empty, with comments indicating where to place logic for clearing the display and preventing leading zeros.

The right sidebar shows the **Diagnostic Tools** window, which includes a **Diagnostics session** (10:15 minutes), a **Process Memory (MB)** graph, and a **Memory Usage** section. The **Events** section shows **Show Events (0 of 0)** and **.NET Counters**.

The bottom status bar indicates the current line is 85, column 26, with the encoding set to `SPC CRLF UTF-8 with BOM`.

Visual Studio Code interface showing the development of a Windows Forms application named "My Calculator". The main editor displays the code for `Form1.vb` in Design view, with the `Click` event handler for `btnEquals` selected. The code includes initialization of variables (`Dim operand1 As Double = 0`, `Dim currentOperator As String = ""`, `Dim isCalculationDone As Boolean = False`) and a `Form_Load` event handler. The `Click` event handler for `btnEquals` is currently empty, with comments indicating where to place logic for clearing the display and preventing leading zeros.

The right sidebar shows the **Diagnostic Tools** window, which includes a **Diagnostics session** (10:57 minutes), a **Process Memory (MB)** graph, and a **Memory Usage** section. The **Events** section shows **Show Events (0 of 0)** and **.NET Counters**.

The bottom status bar indicates the current line is 85, column 26, with the encoding set to `SPC CRLF UTF-8 with BOM`.

Visual Studio Code interface showing the source code of a Windows Forms application named "My Calculator". The code is in VB.NET and is currently in the "Debug" state. The main thread is running at [11168].

```
42 ' Don't allow starting with an operator if display is empty or error
43 If lblDisplay.Text = "0" OrElse lblDisplay.Text.Contains("Error") Then Exit Sub
44
45 ' If an operator is already present in the string, ignore extra clicks
46 If lblDisplay.Text.Contains("+") OrElse lblDisplay.Text.Contains("-") OrElse lblDisplay.Text.Contains("x") OrElse lblDisplay.Text.Contains("/") Then Exit Sub
47 End If
48
49 ' Save the first number and the operator symbol, then append it to the screen text
50 currentOperator = btn.Text.Trim
51 lblDisplay.Text &= " " & currentOperator & " "
52 isCalculationDone = False
53 End Sub
54
55 ' --- 3. EQUALS BUTTON (=) ---
56 ' Double click your '=' button and make sure it points here
57 0 references
58 Private Sub btnEquals_Click(sender As Object, e As EventArgs) Handles btnEquals.Click
59 Try
60 ' Split the display text by spaces to read parts (e.g., "9", "x", "5")
61 Dim parts As String() = lblDisplay.Text.Split(New Char() {" "c}, StringSplitOptions.RemoveEmptyEntries)
62 ' Ensure we have all 3 parts needed for a basic equation
63 If parts.Length < 3 Then Exit Sub
64
```

The right-hand side of the interface shows the "Diagnostic Tools" pane, which includes a "Diagnostics session: 11:37 minutes" and a "Process Memory (MB)" graph. The "Events" pane shows "Show Events (0 of 0)".

Visual Studio Code interface showing the source code of the "My Calculator" application, continuing from the previous screenshot. The code is in VB.NET and is currently in the "Debug" state. The main thread is running at [11168].

```
64 If parts.Length < 3 Then Exit Sub
65
66 operand1 = Convert.ToDouble(parts(0))
67 Dim op As String = parts(1)
68 Dim operand2 As Double = Convert.ToDouble(parts(2))
69 Dim result As Double = 0
70
71 Select Case op
72 Case "+"
73 result = operand1 + operand2
74 Case "-"
75 result = operand1 - operand2
76 Case "x"
77 result = operand1 * operand2
78 Case "/"
79 If operand2 = 0 Then
80 lblDisplay.Text = "Cannot divide by zero"
81 Exit Sub
82 Else
83 result = operand1 / operand2
84 End If
85 Case Else
86 Exit Sub
87 End Select
```

The right-hand side of the interface shows the "Diagnostic Tools" pane, which includes a "Diagnostics session: 12:00 minutes" and a "Process Memory (MB)" graph. The "Events" pane shows "Show Events (0 of 0)".

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search My Calculator

Process: [11336] MyCalculator.exe Lifecycle Events Thread: [11168] Main Thread Stack frame: MyCalculator.Form1.Number_Click

Form1.vb [Design] Form1.vb btnEquals Click

```
88
89 ' This formatting shows "9 x 5 = 45" completely on screen for your assignment snapshot!
90 lblDisplay.Text &= " " & result.ToString()
91 isCalculationDone = True
92
93 Catch ex As Exception
94     lblDisplay.Text = "Error"
95 End Try
96 End Sub
97
98 ' --- 4. CLEAR (C) BUTTON ---
99 0 references
100 Private Sub btnC_Click(sender As Object, e As EventArgs) Handles btnC.Click
101     lblDisplay.Text = "0"
102     operand1 = 0
103     currentOperator = ""
104     isCalculationDone = False
105 End Sub
106 End Class
```

100 % No issues found Ln: 85, Ch: 26 SPC CRLF UTF-8 with BOM

Autos Search (Ctrl+E) Search Depth: 3

Call Stack Search (Ctrl+E) View all Threads Show External Code

Autos Locals Watch 1

Ready 18°C Mostly cloudy 11:09 PM 6/30/2026

Diagnostics Tools

Diagnostics session: 12:46 minutes 12:40min

Add counter graphs by checking counters from counter options

Events

Process Memory (MB) 26 26

Summary Events Counters Memory Usage CP

Events

Show Events (0 of 0)

.NET Counters

View Performance Counters

Memory Usage